

$$x^T V x + 2 v^T x + f = 0$$

$$V = \begin{pmatrix} a & b \\ b & c \end{pmatrix} \quad v = (v_1, v_2)$$

$$\begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + 2(v_1 v_2) \begin{pmatrix} x \\ y \end{pmatrix} + f = 0$$

$$\begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} ax + by \\ bx + cy \end{pmatrix} + 2(v_1 x + v_2 y) + f = 0$$

$$ax^2 + 2bxy + cy^2 + 2(v_1 x + v_2 y) + f = 0$$

$$x^2 - y^2 = 0$$

$$f_1 = 0$$

compare

$$a = 1, \quad v_2 = -1$$

$$V_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$v_2^T = \begin{pmatrix} 0 & -1 \end{pmatrix}$$

$$0 = (1+x) \left(\frac{1}{x} \right)$$

$$x^2 + 2x + y + 1 = 0$$

$$f_2 = 1, a = 1, b = 0, c = 0,$$

$$v_1 = 1, v_2 = \frac{1}{2}$$

$$V_2 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, V_2^T = \begin{pmatrix} 1 & \frac{1}{2} \end{pmatrix}$$

Intersection

$$x^T (v_1 + \mu v_2) x + 2(v_1 + \mu v_2)^T x + (f_1 + \mu f_2) = 0$$

$$(x \ y) \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} (1 + \mu) \begin{pmatrix} x \\ y \end{pmatrix} + 2 \left[\mu \quad -\frac{1}{2} + \frac{\mu}{2} \right] \begin{pmatrix} x \\ y \end{pmatrix} + \mu = 0$$

$$(x \ y) \begin{pmatrix} 1 + \mu & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + 2 \left[\mu \quad \frac{\mu - 1}{2} \right] \begin{pmatrix} x \\ y \end{pmatrix} + \mu = 0$$

$$\rightarrow \begin{vmatrix} v_1 + \mu v_2 & v_1 + \mu v_2 \\ (v_1 + \mu v_2)^T & f_1 + \mu f_2 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 1 + \mu & 0 & \mu \\ 0 & 0 & \frac{\mu - 1}{2} \\ \mu & \frac{\mu - 1}{2} & \mu \end{vmatrix} = 0$$

$$-(1 + \mu) \left(\frac{\mu - 1}{2} \right)^2 + \mu (0) = 0$$

$$\left(\frac{\mu - 1}{2} \right)^2 (\mu + 1) = 0$$

$$\boxed{\mu = 1}$$

$$\boxed{\mu = -1}$$

$$t = k +$$

$$x = h + km$$

Now intersection eqn $\mu = 1, \mu = -1$

$$x^T (v_1 + \mu v_2) x + 2 \left(\begin{bmatrix} 0 \\ -\frac{1}{2} \end{bmatrix} + \mu \begin{bmatrix} 1 \\ \frac{1}{2} \end{bmatrix} \right)^T x + f_1 + \mu f_2 = 0$$

with $\mu = 1 \rightarrow v_1 = v_2 \rightarrow x^2$ term $\rightarrow$ not a chord

$$\mu = -1 \quad 2 \begin{bmatrix} -1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + (-1) = 0$$

$$-2x - 2y - 1 = 0$$

$$\boxed{2x + 2y + 1 = 0}$$